NOTES:

1. EXISTING RINGMAINS WILL SUPPLY TO NEW GRAVITY CIRCUIT IN STAGE 1. NEW DISTRIBUTION SYSTEM WILL REPLACE STAGE 1 IN STAGE 2.
2. FLANGED ANSI B16.5 CL150.
3. REFER TO 2000-F24-PID-PR-10034 FOR GLAND WATER DETAIL.

SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
14-1297-PSW-PE16-32

14-0254-PSW-SS3-50
50x25
25VB03S1 F14-10468
SP MC-001 SP@ TOP OF IRL PLATFORM

SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
14-0224-PSW-PE16-90
PE16 SS2
14-0225-PSW-SS2-80
SS3 SS2
80VCE1S1 F14-10512
SS2 SS3
80VBFD1S2 F14-10513
14-0226-PSW-SS2-50
50VB03S1 F14-10514

GRAVITY CONCENTRATOR 1
2000-F14-PID-PR-10001
GRAVITY CONCENTRATE
14-0207-MS-RL3-250

GRAVITY CONCENTRATOR 2
2000-F14-PID-PR-10001
GRAVITY CONCENTRATE
14-0208-MS-RL3-250

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LEACH AID
(BY HAND)

F14-TK-410 INTENSIVE CYANIDE LEACH REACTOR SOLUTION TANK
F14-PP-421 INTENSIVE CYANIDE LEACH REACTOR DISCHARGE PUMP
F14-HT-414 INTENSIVE CYANIDE LEACH REACTOR SOLUTION TANK HEATER 1
F14-FS-420 INTENSIVE CYANIDE LEACH REACTOR POWDER SCREW
F14-HP-420 GRAVITY CONCENTRATE HOPPER
F14-PP-422 INTENSIVE CYANIDE LEACH REACTOR DRAIN PUMP
F14-HT-415 INTENSIVE CYANIDE LEACH REACTOR SOLUTION TANK HEATER 2
F14-AG-410 INTENSIVE CYANIDE LEACH REACTOR SOLUTION TANK AGITATOR
F14-SA-406 INTENSIVE CYANIDE LEACH REACTOR PREGNANT SOLUTION SAMPLER
F14-PP-423 INTENSIVE CYANIDE LEACH REACTOR SOLUTION PUMP
F14-HT-416 INTENSIVE CYANIDE LEACH REACTOR SOLUTION TANK HEATER 3
F14-IL-403 INTENSIVE CYANIDE LEACH REACTOR

F14-PK-403 INTENSIVE CYANIDE LEACH REACTOR PACKAGE

CYANIDE RINGMAIN
2000-F00-PID-PR-10101
CYANIDE
00-3108-CN-PEDC1-32
BROWNFIELDS 03-02 GREENFIELDS
72-1091-CN-PE8-32
PE8 MS1
25VBD1C1 F14-10519
25VBD1C1 F14-10521
14-0220-CN-MS1-25
NOTE 1
25VBD1C1 F14-10526
25VBD1C1 F14-10528
25VBD1C1 F14-10520
14-0548-CN-MS1-25
TO GRADE
25VBD1C1 F14-10527
14-0931-CN-MS1-25
TO GRADE
FI 14121 H L
FIT 14121
FE 14121

CYANIDE RINGMAIN PUMPS
2000-F00-PID-PR-10102
CYANIDE
2
72-1079-CN-PE16-32
TIP 2088
STAGE 2 STAGE 1
25VBD1C1 F14-10522
25VBD1C1 F14-10523
25VBD1C1 F14-10524
14-2065-CN-MS1-25
14-0938-CN-MS1-25
TO GRADE
FI 14122 H L
FIT 14122
FE 14122

CAUSTIC RINGMAIN
2000-F66-PID-PR-10006
CAUSTIC
14-3036-NS-PE8-32
BROWNFIELDS 03-03 GREENFIELDS
TI
72-1110-NS-PE8-32-TI
PE8 MS1
40x25
40VBD1C1 F14-10529
40VBD1C1 F14-10531
14-0221-NS-MS1-40-TI
NOTE 1
40VBD1C1 F14-10536
40VBD1C1 F14-10538
TI
40x25
40VBD1C1 F14-10530
14-0891-NS-MS1-25
TO GRADE
TI
40VBD1C1 F14-10537
14-0939-NS-MS1-25
TO GRADE

CAUSTIC RINGMAIN PUMPS
2000-F72-PID-PR-10006
CAUSTIC
TIP 2087
STAGE 2 STAGE 1
72-1099-NS-MS1-40-TI
TI
40VBD1C1 F14-10532
25VBD1C1 F14-10533
40VBD1C1 F14-10534
14-2067-NS-MS1-40-TI
14-0965-NS-MS1-25
TO GRADE

RAW WATER PUMPS
2000-F24-PID-PR-10018
RAW WATER
14-0227-BW-PE8-90
PE8 SS1
14-0228-BW-SS1-80
80VBFD1S2 F14-10539

14-0230-IA-MSG3-25
25VB03S1 F14-10540
25x15
DN15 BSP

CLASSIFICATION AIR RECEIVER
2000-F24-PID-PR-10025
PLANT AIR
14-0229-IA-MSG3-25
40x25
25VB03S1 F14-10541
25x15
DN15 BSP

GLAND WATER DISTRIBUTION
2000-F24-PID-PR-10003
GLAND WATER
14-0240-GW-SS3-25
NOTE 3
25x15

DETAIL 7
HS 14074 CYANIDE ADDITION VALVE
DETAIL 7
HS 14075 CAUSTIC ADDITION VALVE
DETAIL 7
HS 14076 LEACH OVERFLOW VALVE
80VBFD1S2 F14-10515
80VBFD1S2 F14-10516
14-0860-PE-SS1-25
25VB03S1 F14-10517
TO SUMP

HS 14083 CONCENTRATE HOPPER DECANT VALVE
HS 14084 CONCENTRATE HOPPER FLUIDISATION VALVE
HS 14085 CONCENTRATE HOPPER FLUIDISATION VALVE
HS 14086 CONCENTRATE HOPPER DISCHARGE VALVE
HS 14087 ACACIA RINSE SPRAY VALVE
HS 14088 DESLIME OVERFLOW VALVE
HS 14089 LEACH DECANT VALVE
HS 14090 WATER DECANT VALVE
HS 14091 ACACIA SLURRY DISCHARGE VALVE

14-0239-PE-SS1-80
100x80
BL GREENFIELDS BROWNFIELDS 05-03
00-3107-PE-PEDC1-110
GRAVITY ELECTROWINNING TANK
2000-F00-PID-PR-10101
PREGNANT SOLUTION

ASHH 14101 0 YA 14102 YA 14102 SIREN & BEACON
ASHH 14101
AIT 14101
AI 14101
AE 14101 HCN GAS MONITOR

25VBE1S1 F14-10518
32x15
SAMPLER PARK POSITION
ZA 14104
SPRAY WATER
HS 14105

M
14-0235-CS-PE10-110
160x110
SAMPLER
O/F
14-0236-CS-PE10-225
BALL MILL DISCHARGE LAUNDER
2000-F13-PID-PR-10002
INTENSIVE CYANIDATION TAILINGS
VENDOR PACKAGE 2000-ME027-E02-10001
14-0943-CS-PE10-90
SAMPLE BUCKET
F14-SA-407
INTENSIVE CYANIDE LEACH REACTOR RESIDUE DISCHARGE SAMPLER

HS 14092 ACACIA FLUSH WATER VALVE
HS 14093 F14-PP-421 SUCTION FLUSHING VALVE
HS 14094 ACACIA DRAIN VALVE
HS 14095 ACACIA FEED VALVE
HS 14106 PREGNANT SOLUTION DISCHARGE VALVE
HS 14107 ACACIA DRAIN WATER VALVE

HS 14103 F14-PP-421 GLAND VALVE
HS 14108 ACACIA DRAIN TO SOLUTION TANK VALVE
HS 14109 PP05 AIR SUPPLY VALVE
HS 14110 ACACIA DESLIME WATER VALVE
HS 14111 ACACIA SOLUTION PUMP DISCHARGE VALVE
HS 14112 ACACIA AUTO SAMPLE VALVE
HS 14115 FRESH WATER ADDITION VALVE

14-0057-FD-PE10-90
BALL MILL DISCHARGE LAUNDER
2000-F13-PID-PR-10002
SPILLAGE
PE10
RL1
14-0056-FD-RL1-80
SP MC-002
50VB03S1 F14-10543
80VKGD1S1 F14-10544
SP MC-002
50VB03S1 F14-10542
LAH 14071
LSH 14071
M
80x65
NOTE 2
F14-PS-408
ILR AREA SUMP PUMP

DN100 14-0231-MS-PE10-110 TO GRADE
DN150 14-0232-PE-SS1-150 TO GRADE
DN100 14-0234-PE-MS1-100 TO GRADE
2
VENDOR PACKAGE 2000-ME052-E02-10001
WI 14073 HH H
WQI 14073
FIC 14079 H L
F14-PP-423 FLOWRATE
PI 14078 HH H
LEACH REACTOR BACK PRESSURE
LI 14077 HH H L
LEACH REACTOR LEVEL
LI 14080 HH H L LL
SOLUTION TANK LEVEL
TI 14081 HH H L
SOLUTION TANK TEMPERATURE
FI 14082 L LL
SOLUTION TANK FLOWRATE
LI 14123 H L LEACHAID HOPPER LEVEL
FAL 14124 LEACHAID AIR PURGE FLOW
TI 14126 HH H LEACHAID SCREW FEEDER TEMPERATURE
HS 14125 LEACHAID AIR PURGE SOLENOID

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
			AH	MM	JBe	BS	DM
2	05SEP25	GENERAL REVISIONS ADDED	AH	MM	JBe	BS	DM
1	10JUL25	ANALOGUE INDICATORS RELOCATED AND GENERAL REVISIONS ADDED	AH	MM	JBe	BS	DM
0	28OCT24	APPROVED FOR CONSTRUCTION	NB	JDM	LS	LS	DM

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: GRAVITY CIRCUIT & INTENSIVE LEACHING
INTENSIVE CYANIDATION - STAGE 1
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F14-PID-PR-10003	2

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